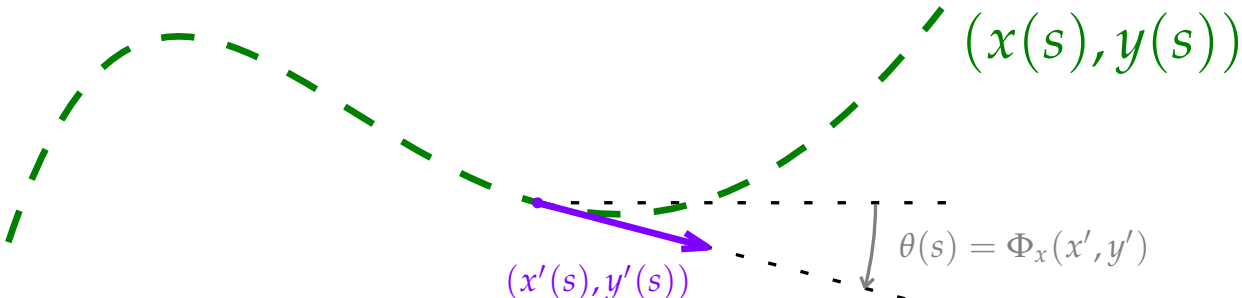
$(x(s), y(s))$
$\theta(s) = \Phi_x(x', y')$
$(x'(s), y'(s))$